

Linear Regression

Technical Notes

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Linear regression is a statistical method used to model the relationship between a response variable and one or more predictors, or covariates. When there is a single predictor, the model is called **Simple Linear Regression**; when there are two or more predictors, it is called **Multiple Linear Regression**.

Linear regression is a fundamental statistical model and is also widely used as a supervised machine learning method for regression problems. It provides a natural starting point for studying regression, introducing many of the concepts that also appear in more sophisticated models.

1 Simple Linear Regression

Suppose we have a training dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N, \tag{1}$$

where x_i is the predictor and y_i is the corresponding response for observation i . In tabular form,

y	x
y_1	x_1
y_2	x_2
$\vdots$	$\vdots$
y_i	x_i
$\vdots$	$\vdots$

y	x
y_n	x_N

Table 1: Training dataset for simple linear regression.

Suppose that the conditional mean of the response can reasonably be modeled as a linear function of the predictor. The simple linear regression model is

$$y_i = \theta_0 + \theta_1 x_i + \varepsilon_i, \quad (2)$$

where θ_0 is the **intercept**, θ_1 is the **slope**, and ε_i is an error term representing the part of y_i that is not explained by the linear relationship.

The systematic component of the model is therefore

$$\mathbb{E}[Y \mid X = x] = \theta_0 + \theta_1 x. \quad (3)$$

The goal is to estimate the unknown parameters θ_0 and θ_1 from the observed data. Once estimates $\hat{\theta}_0$ and $\hat{\theta}_1$ have been obtained, the fitted regression line is

$$\hat{y}_i = \hat{\theta}_0 + \hat{\theta}_1 x_i. \quad (4)$$

For each observation, the difference between the observed response and its fitted value is called the **residual**:

$$e_i = y_i - \hat{y}_i. \quad (5)$$

The question is then how to choose $\hat{\theta}_0$ and $\hat{\theta}_1$ so that the fitted line represents the observed data as well as possible. The standard approach is **Ordinary Least Squares**.

1.1 Ordinary Least Squares

```

1 # Linear regression parameters
2
3 def min_sq(x,y):
4     """
```

Residual Sum of Squares (RSS)

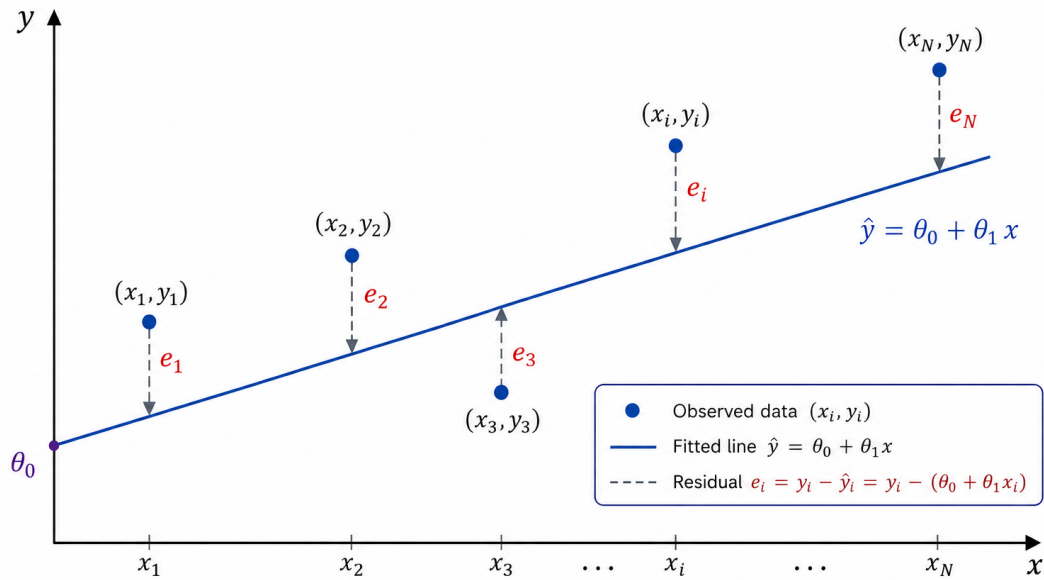


Figure 1: Residuals between the observed responses and the fitted regression line used to compute the Residual Sum of Squares (RSS).

Short Description

Inputs:

x, y

Outputs:

theta_1, theta_0

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```
x_bar, y_bar = np.mean(x), np.mean(y)
theta_1 = np.dot(x - x_bar, y - y_bar) / np.linalg.norm(x - x_bar)**2
theta_0 = y_bar - theta_1 * x_bar
return [theta_1, theta_0]
```